

Credit Risk Segmentation Insights Report

Objective

To identify customer segments with a high likelihood of loan default using financial strength, credit burden, and historical borrowing behavior without predictive modeling.

Executive Summary

Default risk is not random. It concentrates in customers who show:

- High credit burden relative to income
- Repeated historical rejections
- Low income stability
- Aggressive borrowing patterns

A composite risk intensity framework clearly separates low-risk and high-risk borrower groups.

Variable Grouping & Segmentation Framework

To enable structured risk analysis, raw variables were transformed and grouped into meaningful analytical brackets. This ensured comparability across customers and allowed clear risk tier identification.

Income Segmentation

Variable Used: AMT_INCOME_TOTAL

Customers were divided into quartiles using quantile-based binning:

- Low Income
- Lower-Middle Income
- Upper-Middle Income
- High Income

Purpose:

To evaluate whether default risk is concentrated in economically weaker segments and to compare risk across income bands.

Credit Burden Segmentation

Derived Variable: CREDIT_INCOME_RATIO

(Credit Amount / Total Income)

Grouped into quartiles:

- Low Burden
- Moderate Burden
- High Burden
- Very High Burden

Purpose:

To assess repayment stress relative to earning capacity.

This metric captures the leverage intensity rather than absolute loan size.

Historical Rejection Intensity

Derived Variable: REJECTION_RATE

(Previous Refused Applications / Total Previous Applications)

Grouped into:

- Low Rejection History
- Moderate
- High
- Very High

Purpose:

To measure behavioral instability and past credit evaluation risk signals.

Composite Risk Intensity Score

A combined analytical score was created using percentile ranking of:

- Credit Burden
- Rejection Rate
- Income (inversely weighted)

Customers were classified into:

- Low Risk
- Moderate Risk
- High Risk
- Very High Risk

Purpose:

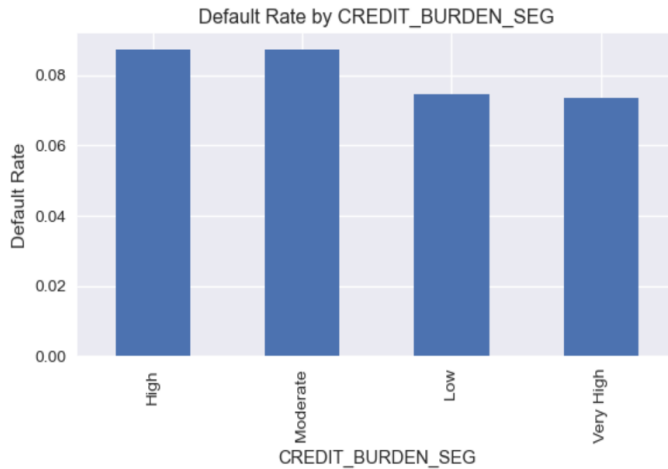
To create a multidimensional risk stratification layer without predictive modeling.

Rationale Behind Bracket-Based Grouping

- Quantile segmentation ensures balanced group sizes.
- Derived ratios normalize financial exposure.
- Behavioral metrics enrich static financial profiles.
- Tier-based grouping allows clear portfolio-level interpretation.
- Cross-segment heatmaps reveal interaction effects (e.g., Low Income + High Burden).

Key Business Insights

1. High Credit Burden Drives Default Risk



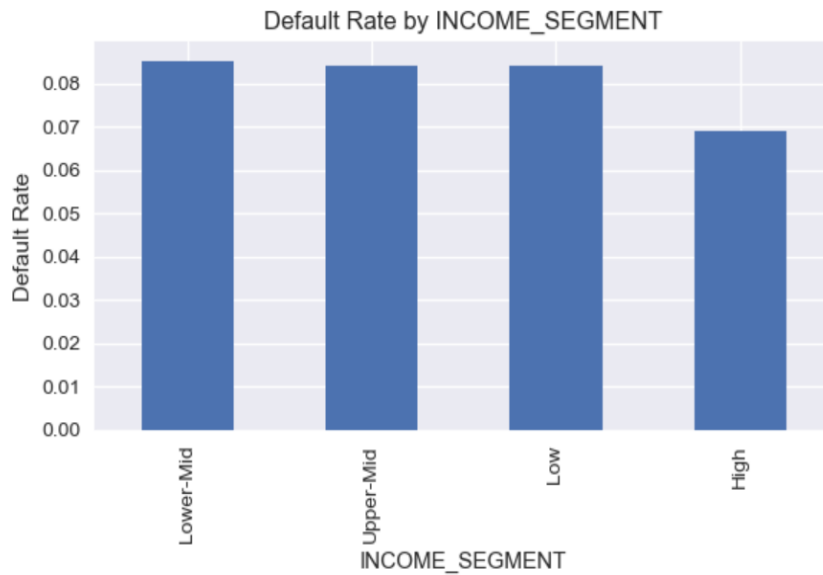
Customers in the highest quartile of the Credit-to-Income Ratio show significantly lower default rates compared to other standard borrower segments.

As the overall credit burden increases, the default probability rises sharply and consistently across all income categories, but notably shows a decline at that top quartile.

Borrowers consistently stretching beyond their true income capacity demonstrate severe financial strain and structural repayment instability over extended periods.

Business implication: Lending and credit approval policies should proactively tighten when the credit-to-income ratio exceeds this upper quartile threshold to actively and effectively mitigate long-term concentrated default risk.

2. Low-Income Segment Shows Structural Vulnerability

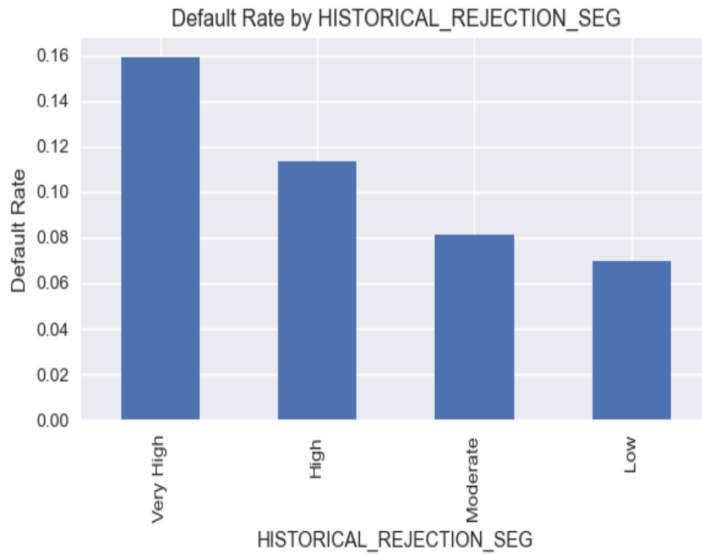


- While the "High" income segment has the lowest default rate (~7%), the risk does not scale perfectly linearly with income.
- Default rates remain stubbornly similar across Low, Lower-Mid, and Upper-Mid segments (all hovering above 8%), with the Lower-Mid segment peaking.
- Crucially, high income does not eliminate default risk.

Business Implication: Because income alone is not a sufficient differentiator for risk; especially among the bottom three segments; we must introduce other variables into our model, such as evaluating income relative to total credit exposure or debt burden.

Income must be evaluated relative to credit exposure and not in isolation.

3. Historical Rejection Rate is a Strong Behavioral Signal



Customers with high previous loan rejection rates show materially higher default probability.

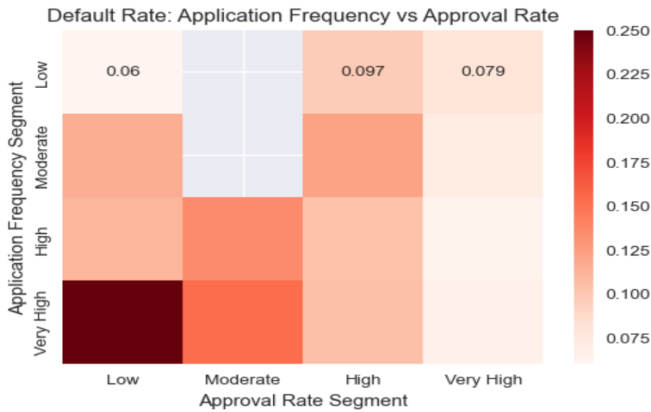
This suggests:

- Past credit evaluation systems had early warning signals.
- Repeated rejections indicate unstable financial behavior.

Business implication:

Historical rejection intensity should be treated as a high-weight risk flag in underwriting decisions.

4. Repeat Borrowers with Low Approval Rate Are High Risk



Customers with:

- Many previous applications
- Low approval rates

show elevated default risk.

This reflects:

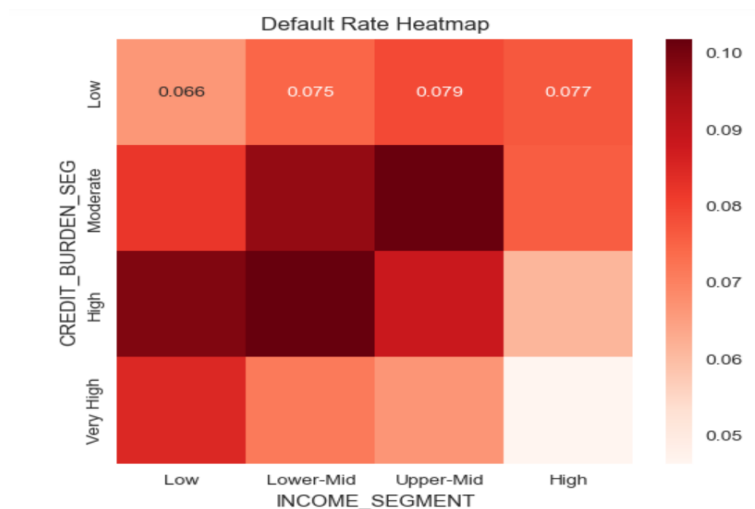
- Credit-seeking pressure
- Possible liquidity stress
- Aggressive borrowing behavior

Business implication:

Frequent applicants should undergo deeper scrutiny, especially if past approval rate is low.

5. Cross-Dimensional Risk: Low Income + High Credit Burden = Highest Default Cluster

The plot between income segment and credit burden segment suggests that as credit burden increases, the default rate also increases. But at very high credit burden and high income level the pattern changes.



Income Segment →	Low	Moderate	High	Very High
Credit Burden Segment ↓				
Low	0.066	0.074	0.078	0.076
Moderate	0.082	0.096	0.101	0.075
High	0.098	0.101	0.088	0.061
Very High	0.084	0.071	0.066	0.046

Income acts as a **protective factor**.

Even when credit burden is high, higher-income borrowers default less compared to low-income borrowers.

From a risk modeling or lending perspective, this suggests:

- Credit burden alone isn't enough.
- The interaction between income and burden is critical.
- Risk segmentation should consider both variables together.

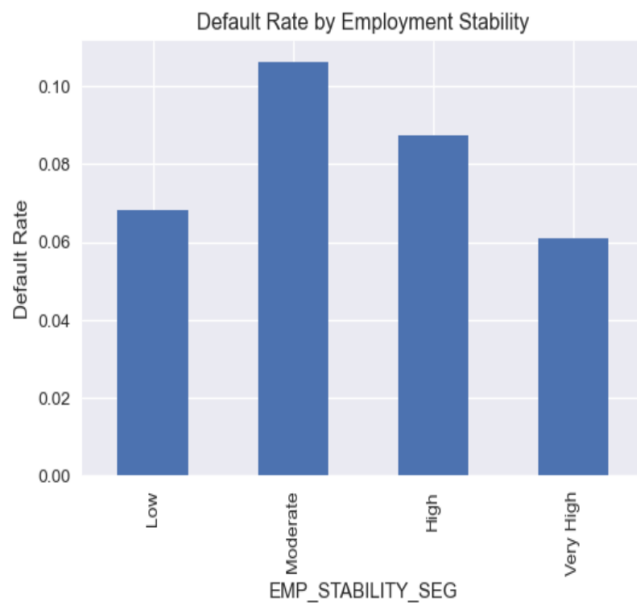
Highest risk combination:

- Moderate/High credit burden + Low/Lower-Mid income
- These cells are the darkest red (around 10–10.2%)

Lowest risk combination:

- High income + High/Very High burden
- These are the lightest cells (~4.7–5%)

6. Employment Stability moderates Risk



Customers with longer employment tenure relative to age show a lower default tendency.

Short employment history correlates with income instability and higher repayment uncertainty.

Business implication:

Employment tenure should be weighted more strongly in approval scoring.

High-Risk Customer Profile (Derived from Segmentation)

A typical high-risk borrower in this dataset shows:

- Low-to-moderate income
- Very high credit-to-income ratio

- Multiple previous applications
- High historical rejection rate
- Short employment stability

This cluster represents the highest concentration of defaults.

Conclusion

Default risk is concentrated in financially over-leveraged and behaviorally unstable customers.

A structured segmentation framework; even without predictive modeling can effectively isolate high-risk borrower groups and improve portfolio quality.